

Material Synthesis 2025 (MatSyn25) Data Set for 2D Materials

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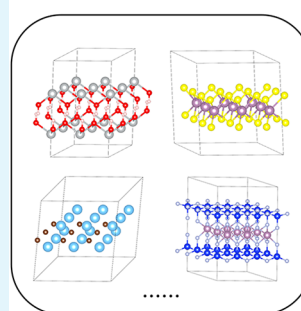


Supporting Information

ABSTRACT: Two-dimensional (2D) materials have shown broad application prospects in fields such as energy, environment, and aerospace owing to their unique electrical, mechanical, thermal, and other properties. With the development of artificial intelligence (AI), the discovery and design of novel 2D materials have been significantly accelerated. However, due to the lack of basic theories of material synthesis, identifying reliable synthesis processes for theoretically designed materials is a challenge. The emergence of large language model offers approaches for the reliability prediction of material synthesis processes. However, its development is limited by the lack of publicly available data sets of material synthesis processes. To address this, we present the Material Synthesis 2025 (MatSyn25), a large-scale open data set of 2D material synthesis processes. MatSyn25 contains 163,240 pieces of synthesis process information extracted from 85,160 high-quality research articles, each including basic material information and detailed synthesis process steps. Based on MatSyn25, we developed MatSyn AI, which specializes in material synthesis, and provided an interactive web platform that enables multifaceted exploration of the data set (<https://matsynai.stpaper.cn/>). MatSyn25 is publicly available, allowing the research community to build upon our work and further advance AI-assisted materials science.

KEYWORDS: material synthesis, two-dimensional materials, scientific data set, large language model, interactive platform

Material Synthesis 2025 (MatSyn25)



Open source data

Web platform
 MatSyn AI85160
Journal articles182299
2D Materials163240
Synthesis processes

INTRODUCTION

The discovery of novel materials is fundamental to addressing many pressing global challenges. This includes two-dimensional (2D) materials represented by graphene.^{1–3} Over the past few decades, various 2D materials with diverse elemental compositions and structures have been discovered.^{4–6} Owing to their subnanometer thickness, unique quantum effects, and boundary conditions, they exhibit excellent electrical, optical, mechanical, thermal, and other properties, demonstrating immense potential in fields such as quantum materials, sensors, catalysts, and energy storage.^{7–10} In recent years, with the development of artificial intelligence (AI), inverse design models,¹¹ high-throughput screening technologies based on machine learning (ML) and density functional theory property prediction,^{12–14} and AI-driven autonomous laboratory^{15,16} have significantly accelerated the discovery of novel materials. For instance, reinforcement learning algorithm has accelerated the development of novel 2D materials for perovskite solar cells, significantly improving the carrier transport performance of 2D perovskites by designing novel ammonium ligands.¹⁷ Unsupervised ML framework has discovered novel antisite Te adatom defects and designed hydrogen evolution catalysts with excellent catalytic activity.¹⁸ Furthermore, AI-controlled robot system has achieved the automated synthesis of 2D materials using chemical vapor deposition.¹⁹ However, although AI has greatly enhanced the efficiency of material design and

screening, how to synthesize these theoretically feasible new 2D materials is a difficult problem.

For the synthesis of new 2D materials, a relatively comprehensive and systematic theoretical framework has been established in the field of inorganic materials synthesis. This includes phase formation theory based on phase diagrams and thermodynamic equilibrium conditions,²⁰ solid phase reaction kinetics based on diffusion control mechanisms, and crystal nucleation and growth theory.²¹ These theories have provided an important mechanistic foundation for understanding the energy change, nucleation behavior, phase transformation pathways, and grain evolution of material formation.²² Based on these theories, researchers can predict the possible stable structural intervals of materials, assess the thermodynamic feasibility of reactions,²³ and explain phenomena in solid-state synthesis processes such as sintering, phase transition, and morphology evolution.²⁴ However, although these theories offer principled guidance for understanding

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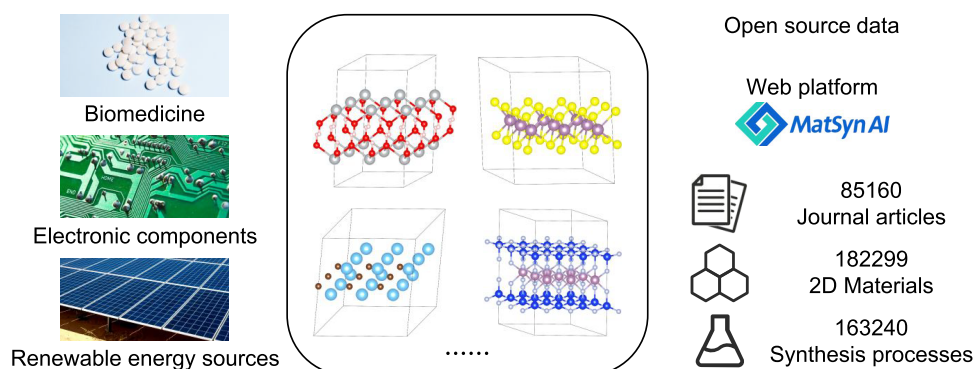


Figure 1. Overview of the materials, application domains, and synthesis process data contained in the MatSyn25 data set.

materials synthesis, they are often based on assumptions such as idealized equilibrium conditions, homogeneous diffusion fields, or specific interface morphologies.²⁵ It is difficult for these theories to precisely provide specific and feasible synthesis steps and optimal process parameters for new 2D materials. The real synthesis process involves numerous adjustable parameters, such as temperature, reaction time, and precursors, forming a complex high-dimensional parameter space. Finding a stable and reproducible synthesis process for theoretically designed novel 2D materials in this space is a huge challenge, often requiring researchers to spend months or even years conducting repetitive experiments and trial and error.^{26,27} Therefore, developing new methods that can assess the synthesizability of designed 2D materials and identify the optimal synthesis process is crucial for accelerating the development of 2D materials.²⁸

In recent years, the emergence of large language models (LLMs), exemplified by ChatGPT, has provided new approaches for the design and optimization of material synthesis processes. With its powerful text understanding and generation capabilities, LLM can effectively process and understand unstructured, text-intensive data such as material synthesis processes, which contain various information forms, including chemical symbols, numbers, and natural language. It has been applied to recommend and optimize synthetic conditions in materials science.²⁹ For instance, Mo et al. developed an LLM-based reaction development framework, which successfully optimized and scaled up the synthesis process for copper/TEMPO catalyzed aerobic oxidation of alcohols to aldehydes, demonstrating the great potential of this technology in specific reaction systems.³⁰ However, the performance and applicability of such models heavily depend on the training data. Therefore, the key foundation for building an LLM specialized in the synthesis process of 2D materials lies in the construction of a high-quality data set of the synthesis process of 2D materials used for model training.

Currently, several studies have attempted to construct material synthesis processes databases through literature mining. For instance, Park et al. employed text mining and natural language processing techniques to extract 33,343 inorganic material synthesis processes from 24,304 materials science papers, providing data support for subsequent retrosynthesis planning work.³¹ Similarly, Omar M. Yaghi team extracted approximately 800 unique metal organic frameworks (MOFs) and 26,257 synthesis parameters from peer-reviewed related research articles by using their self-

developed extraction tool.³² However, these works primarily remain in extracting unstructured descriptive paragraphs or vague operation steps, without systematically parsing and structuring key elements of the synthesis process, such as precursors, reaction conditions, and experimental equipment. MOFs and 2D materials exhibit significant differences in intrinsic properties and corresponding synthesis processes, which further makes it difficult to expand the extraction models and extracted data in different material systems. Moreover, the scale of the extracted data is relatively limited, particularly in the field of 2D materials, where there remains a lack of large-scale, high-quality, and systematic synthesis process data set at present. Therefore, the construction of a high-quality and structured synthesis process data set for 2D materials is not only an essential infrastructure building task to advance intelligent research and development in this domain, but also a core task that urgently needs to be broken through at present.

In this paper, we introduce the Material Synthesis 2025 (MatSyn25) 2D material synthesis process data set, an interactive web platform for accessing this data set, and the MatSyn AI, a large language model fine-tuned on this data set for synthesis process (Figure 1). The MatSyn25 data set contains 163,240 pieces of synthesis process information extracted from 85,160 high-quality papers on 2D materials through our framework specifically developed for automated extraction of synthesis process information. Each piece includes the basic material information and detailed synthetic process steps. The MatSyn25 data set supports multiple channels of interactive exploration and data mining. The developed MatSyn AI, as an intelligent research assistant, can understand queries about material synthesis and recommend synthesis processes that contain information such as precursors, reaction conditions, and equipment. In addition, we developed an open information query platform accessible at <https://matsynai.stpaper.cn/>, where users can retrieve synthesis process information using search tools or interact with MatSyn AI to obtain synthesis process recommendations. The data set is open-sourced (<https://github.com/MatSynAI/MatSyn25>). We hope that this data set can promote the rapid development of AI and materials science, enabling us to take an important step toward developing intelligence-driven methods to understand the synthesis of inorganic materials.

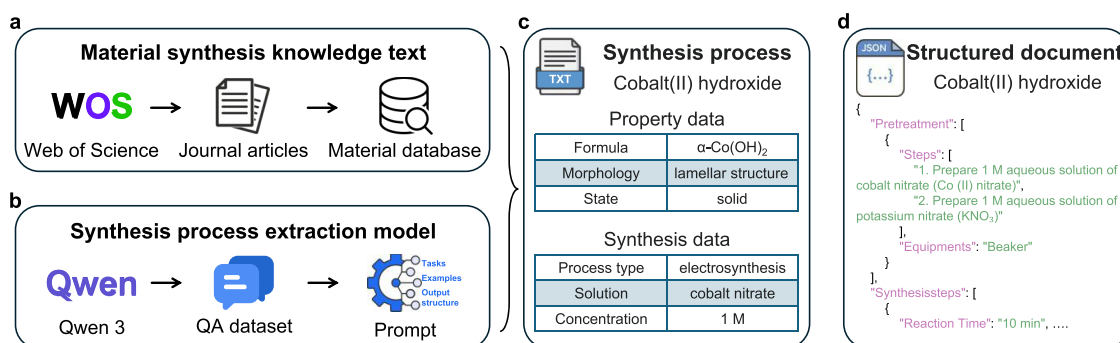


Figure 2. Overall process of the MatSyn25 data set building. (a) Extraction of material synthesis knowledge text. (b) Construction of material synthesis process extraction model. (c) Example of property data and synthesis data of an extracted material. (d) Example of the structured document of an extracted synthetic process.

RESULTS

MatSyn25 Data Set

The MatSyn25 data set is constructed for 2D materials such as graphene, layered double hydroxides (LDH), two-dimensional transition metal carbides and nitrides (MXenes), transition metal dichalcogenides (TMDs), and MoSi_2N_4 . A total of 85,160 relevant research papers were collected, from which detailed synthesis process information for these materials was extracted. MatSyn25 exhibits broad distributions in material types, synthesis processes, and experimental conditions, providing support for systematic research on 2D material synthesis, process optimization, and the development of platforms for novel material discovery.

Figure 2 shows the overall build process of the MatSyn25 Data set. First, the retrieved and downloaded article underwent data cleaning and text conversion, followed by retrieval augmentation based on a materials science knowledge base to construct enhanced knowledge text (Figure 2a). Subsequently, the Qwen3–8B model was fine-tuned using a domain-specific instruction tuning data set to construct a material synthesis process extraction model. Finally, a data extraction prompt was designed, and the fine-tuned model was employed to extract material synthesis processes (Figure 2b).

In this work, a total of 85,160 literatures were collected. The PDF format files were processed using MinerU for structured parsing to extract core content, including text paragraphs and figure/table information.³³ For text format files, regular expressions were applied to remove metadata and format elements that are irrelevant to the research content, such as special formats, characters, headers and footers, and references and other irrelevant information. For image format files, OCR tools were used for text recognition and format conversion. Finally, all processed text content, along with literature metadata (such as DOI, title, abstract, and authors, etc.), was structurally stored in the local material synthesis knowledge text database in JSON format.

Owing to the problems of fragmented information, weak contextual relationships, and information missing in the preprocessed material synthesis knowledge text, knowledge enhancement process is applied to improve the completeness, connectivity, and usability of the knowledge, thereby increasing the efficiency and accuracy of subsequent synthesis process extraction. First, the knowledge text database undergoes context-aware semantic chunking, dividing the material synthesis text into fine-grained technical semantic units. These units are then encoded into high-dimensional semantic

vector representations using a domain-enhanced word embedding model, forming a semantic vector database. Subsequently, based on vector similarity matching, semantic units are retrieved in a preconstructed materials science knowledge base. The retrieved results are used for semantic alignment, information integration, and adding relevant background information, such as material properties and typical process constraints. This process ultimately generates structured enhanced knowledge texts with significantly enriched information. Details of the knowledge enhancement step are provided in Note S1.

With recent advancements demonstrating the outstanding capabilities of LLMs in text understanding and generation, research on using LLMs for information extraction has significantly increased. Compared to traditional approaches relying on manual rules or conventional machine learning methods for structured information extraction from text, LLMs exhibit distinct advantages through their thorough comprehension of complex contextual relationships. They can efficiently extract subtle information from text sources and perform highly complex data extraction tasks. Moreover, LLMs demonstrate significantly less performance degradation than traditional approaches when handling large-scale patterns involving millions of entities, showing superior scalability and practicality. This provides an effective technical pathway for automatically constructing high-quality structured knowledge from massive unstructured texts.^{34–36} The feasibility of this approach has been validated in some domains. For instance, Kim et al. have successfully used LLM to extract MOFs synthesis conditions and property data from more than 40,000 research articles.³⁷ The extraction of 2D material synthesis processes is also a typical task with a large amount of data, high complexity, and multitask coupling. Therefore, in this work, by fine-tuning the pretrained language model, an information extraction model for material synthesis processes was constructed, aiming to efficiently and automatically process material synthesis knowledge texts and extract material synthesis processes. To optimize performance, we have designed task-specific prompts that guide the model in executing material synthesis process extraction with enhanced precision and reliability.

To construct an efficient model for extracting material synthesis process, this study took the pretrained language model Qwen3–8B³⁸ as the base and applied Low-Rank Adaptation³⁹ for fine-tuning. We constructed structured instruction tuning data set related to the extraction task from professional databases and manually labeled corpora as training

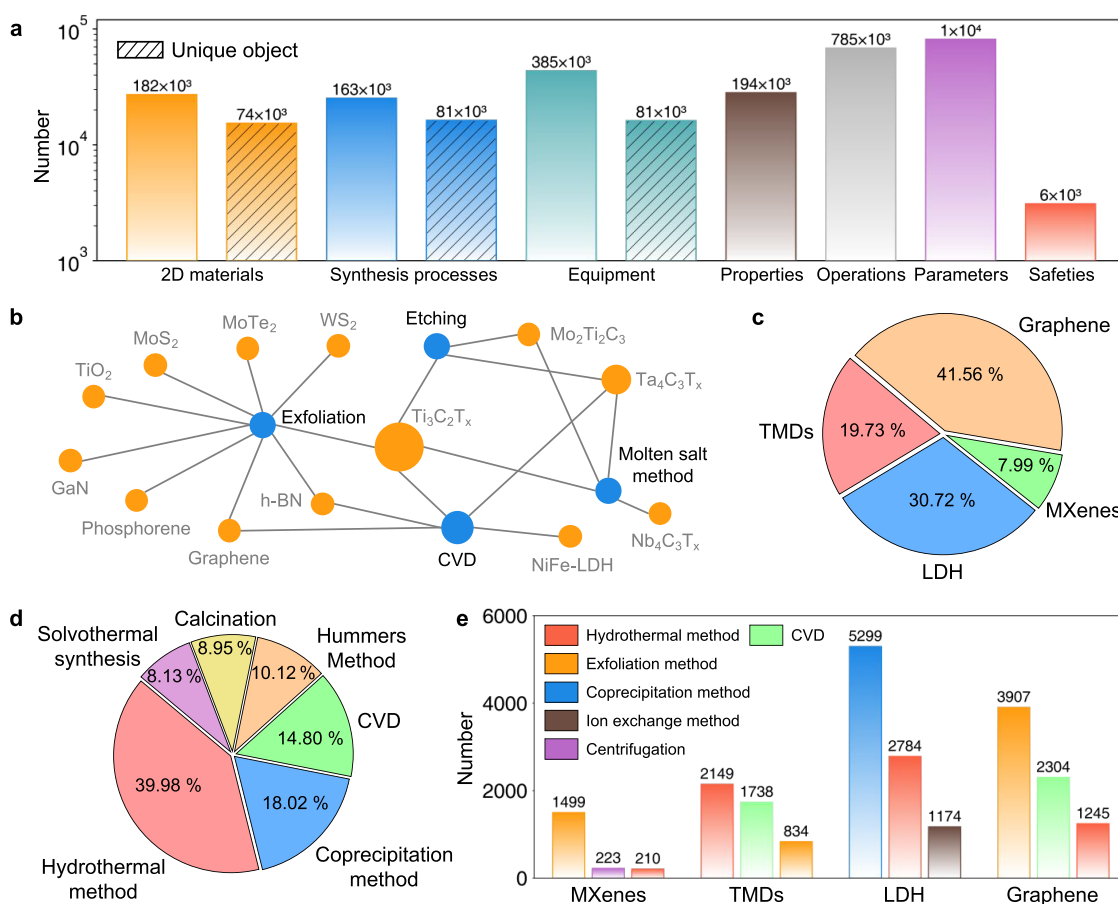


Figure 3. Statistical Analysis of the MatSyn25 Data set. (a) Number of different types of extracted information in the Data set. (b) Representative excerpt of the knowledge graph constructed based on the data set. (c) Distribution of the top four most extracted material types. (d) Distribution of the top six most extracted synthesis process types. (e) Distribution of material types and synthesis process types.

data. Meanwhile, combining task-oriented prompt engineering, a four-layer guidance system was designed, including role definition, feature description, special data processing, and JSON format output constraints, to enhance the model's understanding of the material synthesis. During the fine-tuning process, instruction-response pairs and designed prompts guided the model to learn the extraction logic of synthesis process. This approach only needs to optimize a small number of low-rank parameters, significantly reducing computational costs while enhancing the model's semantic understanding and structured output capabilities through domain knowledge. After performance evaluation and hyperparameter tuning, the model achieved a precision of 0.98 and a recall of 0.95 in the extraction task. More extraction results analysis is provided in Note S2. Ultimately, we deployed the extraction model to automate extract synthesis process from knowledge texts.

To achieve precise extraction of material synthesis process and property information, this study designed structured data extraction prompts to drive the extraction model to perform hierarchical information extraction tasks. The prompts closely revolve around the core elements of material synthesis processes, clarifying key extraction points, including material information, reaction conditions, equipment information, and operation steps. To assist the model in accurately understanding these elements, the prompts contain clear definitions of entity concepts and typical example explanations. Based on this design, we constructed a three-layer progressive prompt structure. First, clearly state the basic task objective, that is,

require the model to identify and extract specific information from the texts. Second, define in detail the connotations of the concepts to be extracted, their related attribute features and possible expressions in the text, and strictly stipulate the data type, format, and overall structure of the output information. Finally, provide a complete output structure example and clarify the processing rules for special data at the same time. By integrating task instructions, entity concept definitions with examples, and structured output constraints, logically rigorous and clearly instructed structured prompts were formed. This ensures the model can accurately identify and extract the required material synthesis process information.

The extraction process adopts a progressive three phase strategy to achieve gradual refinement and completion of the extracted data. First, the extraction model extracts core information from the knowledge texts, including the synthesis process name, type, purpose, as well as involved materials and equipment. Then, the extracted detailed operation steps are divided into three phases: preprocessing, synthesis, and postprocessing. Subsequently, based on the operation steps of each phase, the model performs in-depth mining of the associated attributes within each step, structurally linking the specific details and reaction conditions during operation to the corresponding substeps. Finally, for possible missing information, precise identification and supplementation are carried out by matching the extracted information in the external professional database. Details of the information supplementation step are provided in Note S3.

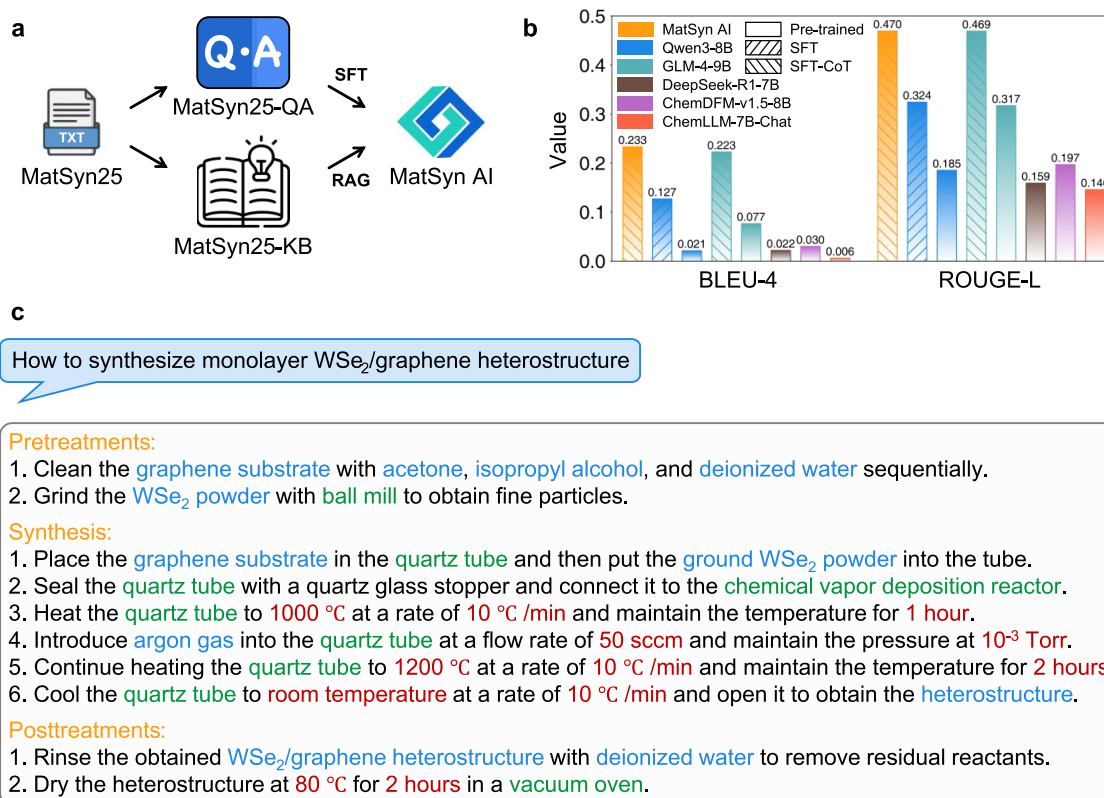


Figure 4. Model architecture and performance of MatSyn AI. (a) Model architecture of MatSyn AI. (b) Performance evaluation of MatSyn AI. (c) An example of MatSyn AI's response, which includes the process stage (yellow), the required reagents and materials (blue), the experimental equipment used (green) and the specific process parameters (red).

To ensure the reliability of the extraction results, we conducted multiple rounds of expert sampling validation before large-scale automated extraction to identify problems existing in the workflow. Based on the feedback from experts, we optimized the extraction process and model prompts. For instance, in the data postprocessing stage, we added unit standardization and conversion module that had not been considered previously. Ultimately, through large-scale extraction, a high-quality, structurally standardized, and comprehensive data set for material synthesis process was constructed.

The complete data set is accessible via <https://github.com/MatSynAI/MatSyn25> and <https://huggingface.co/datasets/MatSynAI/MatSyn25>. Each record corresponds to the comprehensive data for a material system, including the DOI of the paper from which the material was extracted, basic material information, physicochemical properties, experiments safety precautions, precursor materials used in synthesis, as well as detailed operation steps, conditions, and parameter details, etc. (Figure 2c, d)

From collected research papers, we extracted 182,299 2D materials, encompassing 193,633 pieces of physicochemical properties and 163,240 material synthesis processes. Among them, 74,464 kinds of 2D materials are unique. Regarding synthesis processes, we extracted 385,059 pieces of equipment information used in material synthesis, 784,863 pieces of specific operation step information, 1,405,771 specific process parameters (including detailed information such as reaction time, temperature, and reactant quantities), and 5937 safety precautions during experiments (Figure 3a). Based on the extracted information, we constructed a knowledge graph (Figure 3b). This graph clearly demonstrates the intercon-

nections among data points, indicating that the extracted data are not isolated fragments but rather associated data sets that can support reasoning chains and in-depth research. Figure 3c shows the distribution of the top four most types of 2D materials among all extracted materials, with graphene being the most extensively studied. Similarly, Figure 3d shows the distribution of the top six most types of synthesis processes extracted. Among all types, the hydrothermal method is the most commonly used for synthesizing 2D materials. To further analyze the relationship between different 2D materials and their synthesis methods, we visualized the frequency of synthesis methods used for each material (Figure 3e). Among the materials, the hydrothermal method and exfoliation method are the most commonly used methods. More statistical results of the extracted information can be found in the Tables S1–S5.

MatSyn AI

Based on the MatSyn25 data set, we trained a large language model, MatSyn AI, specialized in material synthesis processes (Figure 4a). By leveraging the knowledge of material synthesis processes contained in the data set, this model learns the complex mapping relationships between material properties and synthesis processes. It is capable of automatically recommending synthesis processes for target materials and optimizing synthesis processes.

The construction of a high-quality, structured instruction tuning data set from the MatSyn25 data set is a key prerequisite for training large language models for material synthesis processes. We first screened out 12,317 valid samples from the original 163,240 pieces of data, ensuring that each

sample had a clearly defined target product and complete core synthesis steps, including process flow, parameters, and equipment information. During the question construction phase, we designed diverse question templates covering 8 template-based types, 34 direct-question types, and 6 information-prompt types. Question template examples can be found in the [Note S4](#). For each data sample, one template was randomly selected and filled with corresponding information to generate a question. Additionally, we used the Qwen3–8B model to generate a conditional constraint question for each sample. Finally, each sample corresponds to two questions that are semantically matched but from different perspectives. In the answer construction phase, we adopted a dual-module design consisting of “synthesis process” and “reasoning chain”. The “synthesis process” section directly used the synthesis step text from the original data set. The “reasoning chain” section is reasoning corpus generated by the teacher LLM (DeepSeek-R1⁴⁰) based on all data with similar synthesis processes, by mining the potential relationships between physical properties and synthesis information. This approach effectively establishes explicit reasoning chains from material properties to synthesis processes by leveraging similar material data, providing structured supervisory signals for the model to learn complex decision-making processes. Ultimately, we constructed the MatSyn25-QA data set, comprising 22,234 high-quality question-answer pairs, for fine-tuning the material synthesis process model.

Based on the MatSyn25-QA data set, we used the LLaMA-Factory training framework to conduct supervised fine-tuning (SFT) for constructing specialized model for material synthesis process. To comprehensively evaluate the impact of different model architectures and data, we fine-tuned two base models: Qwen3–8B and GLM-4–9B. For Qwen3–8B, we used two different fine-tuning methods. One approach was to construct Qwen3–8B-SFT using only synthesis process data for SFT. Another approach was to fine-tune the MatSyn AI model by using synthesis process data and chain-of-thought (CoT) reasoning data. Meanwhile, on the GLM-4–9B base model, we also performed fine-tuning using CoT data and synthesis process data, thereby constructing the GLM-4–9B-SFT-CoT model for subsequent comparative analysis. Evaluation on the test set demonstrated that CoT-enhanced model, MatSyn AI, achieved outstanding performance. Its BLEU-4 and ROUGE-L scores are 0.233 and 0.470, respectively, which are superior to other models such as GLM-4–9B-SFT-CoT, DeepSeek-R1–7B, as well as domain models ChemDFM-v1.5–8B⁴¹ and ChemLLM-7B-Chat⁴² ([Figure 4b](#)). Details of model hyperparameter selection and test are provided in [Tables S6–S8 and Notes S5–S6](#). The results of the ablation experiment revealed the influence of data on the model performance. Under SFT without CoT, Qwen3–8B-SFT’s ROUGE-L score increased from 0.185 to 0.324. MatSyn AI, trained with CoT, achieved the highest score of 0.470. This demonstrates that the introduction of reasoning chain significantly enhances model performance.

To enhance the accuracy, reliability, and detail richness of the content generated by MatSyn AI, as well as to strengthen its ability to resist hallucinations when handling complex or rare problems, this study constructed a Retrieval-Augmented Generation (RAG)⁴³ knowledge base, which is named MatSyn25-KB. MatSyn25-KB is based on the MatSyn25 data set and constructed using text generation templates. These templates effectively identify and extract key information from

fragmented data such as process parameters and equipment attributes, and through semantic understanding, contextual reasoning, and logical integration, transform this information into natural language text with clear structure, strict logic, and complete information. Subsequently, an efficient vector retrieval index library was constructed based on these high-quality segments. During the generation phase, the model first uses the embeddings generated by the Qwen3-Embedding model and the precalculated cosine similarity matrix to retrieve the top-5 candidate knowledge segments with the highest semantic similarity to the user’s query. To avoid reasoning chain deviations caused by misjudgment of similarity, an LLM evaluates the relevance of these candidate segments from multiple dimensions, such as rationality and path coherence, and then filters out segments that are superficially similar but essentially irrelevant. Then, MatSyn AI takes these retrieved detailed knowledge segments as context and generates a structurally complete response that conforms to the laws of materials science. Details of the generation phase are provided in [Note S7](#).

To systematically evaluate the effectiveness of this method, we conducted a hallucination assessment experiment. We adopted scoring criteria (on a scale of 1 to 5) to evaluate model outputs from three dimensions: factual similarity (the consistency between the synthesis method output by the model and the standard method), method feasibility (whether the synthesis method output is physically and chemically reasonable), and instruction adherence (whether the model’s response aligns with the query). The test results demonstrated that the Qwen3–8B base model performed poorly in all evaluation metrics. Qwen3–8B-SFT, which was fine-tuned with synthesis process data, achieved foundational improvement in instruction adherence. When the CoT reasoning data was further introduced for fine-tuning, MatSyn AI enhanced feasibility and instruction adherence by reducing logical leaps and omissions through explicit stepwise reasoning. Finally, with the integration of RAG, the output of MatSyn AI became more accurate and reliable. The RAG mechanism ensures that generated content is based on verified synthesis processes retrieved from the MatSyn25-KB knowledge base, thereby mitigating model hallucination and reducing outputs of unverified or fictional processes. Detailed model scores are provided in [Table S9](#). This approach significantly improves the detail completeness and reliability of the response, effectively reduces the risk of model hallucinations, and makes the results more practical. [Figure 4c](#) shows an example of MatSyn AI’s answer to a question. For the query of the material synthesis process, MatSyn AI will answer the detailed operation steps of the synthesis process, including all experimental equipment, required reagents, and specific process parameters that are needed.

To validate the feasibility of the synthesis process recommended by MatSyn AI, an experimental case was designed for “One-step electrodeposition synthesis of coupled Ni nanocrystals and Ni(OH)₂ materials.” Ni(OH)₂ is known as an active phase for oxygen evolution reaction catalysis, but its intrinsic electrical conductivity is relatively low. The introduction of metallic nickel nanocrystals can enhance the conductivity of the coupled material. However, the overlapping area of the thermodynamically stable pH-potential regions of metallic nickel and nickel hydroxide is small.⁴⁴ Therefore, achieving the one-step electrodeposition synthesis of coupled materials requires the model to control appropriate reduction

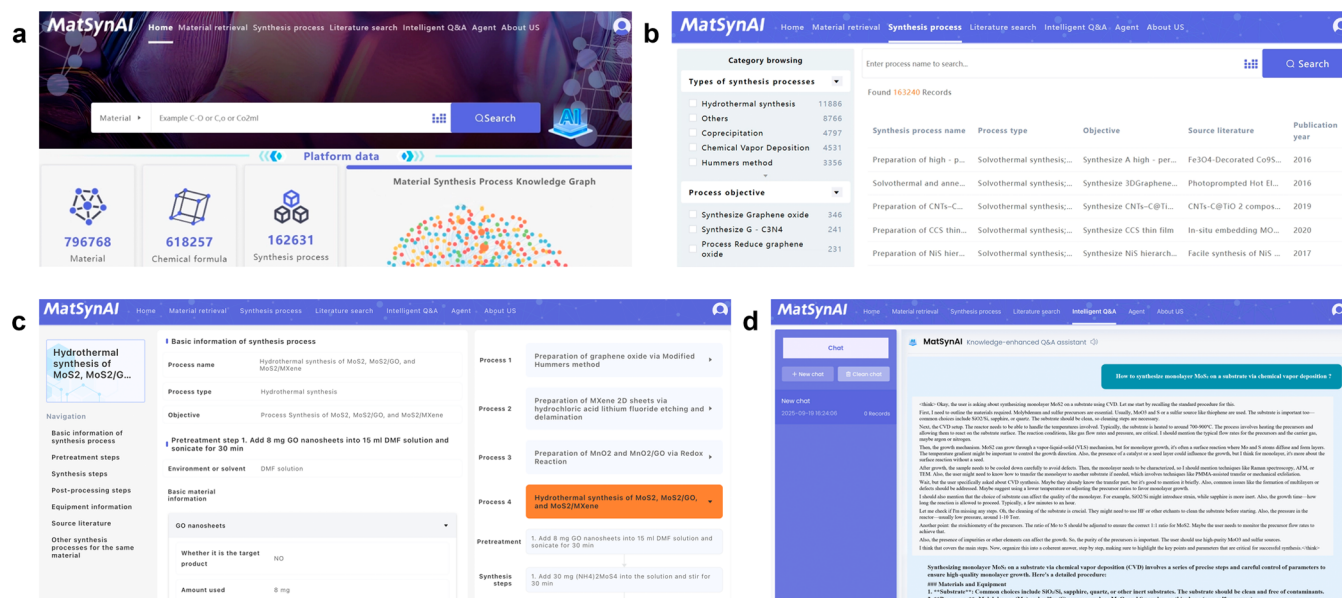


Figure 5. Web platform and detailed information pages. (a) Homepage of the web platform. (b) Search results page. (c) Detailed information page. (d) Question answering system page.

potential to ensure the nucleation and growth of nickel metal nanocrystals, while maintaining a specific electrolyte environment to promote the deposition of $\text{Ni}(\text{OH})_2$. Based on this, we input a query to MatSyn AI: “One-step electrodeposition synthesis of coupled Ni nanocrystals and $\text{Ni}(\text{OH})_2$ materials.” The model successfully generated a complete and operational synthesis process (full details and parameters are provided in Note S8). We strictly followed this process flow to prepare the material. To confirm the phase composition and chemical state of the synthesized material, we performed characterization including X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and Raman spectra. The XRD pattern showed distinct diffraction peaks of $\text{Ni}(111)$ at $2\theta \approx 44.5^\circ$, 51.8° , and 76.4° and the amorphous hump peak of $\text{Ni}(\text{OH})_2$ around $20\text{--}30^\circ$, indicating the presence of Ni nanocrystals and $\text{Ni}(\text{OH})_2$ (Figure S1). XPS analysis further revealed the surface chemical states of as-prepared sample. For the high-resolution Ni 2p spectra, peaks at 855.4 and 873.0 eV were attributed to Ni 2p_{3/2} and Ni 2p_{1/2} of $\text{Ni}^{2+}\text{--O}$, as well as peaks at 851.7 and 868.8 eV were assigned to Ni^0 2p_{3/2} and Ni^0 2p_{1/2} binding energies (Figure S2). These results provide evidence for the presence of both Ni nanocrystals and $\text{Ni}(\text{OH})_2$. Additionally, the Raman spectra displayed characteristic peaks at 445 and 577 cm^{-1} , which are assigned to the $\nu(\text{Ni}^{2+}\text{--O})$ vibrational mode in $\text{Ni}(\text{OH})_2$, further confirming the formation of the $\text{Ni}(\text{OH})_2$ phase (Figure S3). These characterization results confirm the successful preparation of the target coupling material. This case study demonstrates that MatSyn AI can generate experimentally feasible and effective synthesis process. This enhances its practical value as an auxiliary tool in materials synthesis research.

MatSyn Web Platform

Based on the MatSyn25 data set and MatSyn AI, we provide a user-friendly web platform. Users can use search tools to retrieve material and synthesis information, as well as interact with MatSyn AI to obtain synthesis process recommendations (Figure 5a).

The web platform offers three search modes: material search, synthesis process search, and literature search. Material search allows users to query materials by name, chemical formula, or elemental composition. The system will return a list of materials sorted by relevance. Users can further filter the results based on criteria such as material type and morphology. Each search result contains comprehensive material information, including basic properties, detailed synthesis process parameters, operation steps, experimental instrument information, as well as abstract and link to the source literature. Synthesis process search enables users to retrieve by name and filter results by process type, target material, or equipment used (Figure 5b). Each search result will provide detailed information about the synthesis process, abstract, and link to the source literature, and a list of related materials synthesized using this process in the database (Figure 5c). Literature search allows users to retrieve based on material information or keywords. The literature details page provides complete literature metadata and basic information about the synthesis processes and materials extracted from it. All extracted information is linked to their respective detail pages for further exploration.

The web platform also deeply integrates an intelligent question answering system, leveraging artificial intelligence technology to provide comprehensive research support. Users can input professional questions about synthesis paths, parameter optimization, and more into the intelligent question answering interface. The system will invoke MatSyn AI and generate clear and information-rich responses based on the entire database (Figure 5d). For example, for the synthesis requirements of specific 2D material, the system can quickly integrate synthesis methods from multiple literatures, generate structured synthesis processes with detailed parameters, and provide corresponding references for traceability. Additionally, the system offers literature-assisted reading function: users can select literature from the system database or upload their own literature. The system will automatically summarize the literature and extract key information, and support users to ask in-depth questions about the content. The system will

answer based on the original literature and the database. It is worth noting that when questions involve specialized data that has not been included in the database, the system supports users in uploading relevant literature or data as supplementary context. MatSyn AI will conduct comprehensive analysis and reasoning based on this information to provide well-founded answers. This extends the system's applicability and practical flexibility. Furthermore, the web platform is embedded with a global AI assistant that can be activated at any interface, providing users with immediate and accurate knowledge acquisition and problem-solving. The assistant can provide precise answers based on the content currently viewed by the user and in combination with the system database. By combining dedicated interaction with global assistance, the intelligent question answering system provides users with support throughout the entire research workflow, significantly enhancing users' information acquisition efficiency and scientific productivity.

CONCLUSIONS

The MatSyn25 data set is an open access knowledge base dedicated to the systematic integration of synthesis processes for 2D materials. It provides the most comprehensive data on the synthesis processes of 2D materials to date and will be dynamically updated by continuously adding the latest literature. Based on the MatSyn25 data set, we trained MatSyn AI specialized in synthesis processes. This model can generate material synthesis process based on the MatSyn25 data set and analyze and optimize existing processes, providing researchers a powerful tool for synthesis process design and improvement. To promote practical application of data set and model, we integrated them into a user-friendly, open web platform that supports efficient data retrieval and question answering, significantly enhancing the accessibility of the database. MatSyn25 fills the gap in the field of 2D material synthesis process and establishes a foundation for data-driven synthesis research. Its trinity architecture of data set, model, and platform provides essential infrastructure and data resources to accelerate novel material discovery, optimize synthesis process, and advance materials science, demonstrating profound academic influence and application potential.

Looking ahead, future work will focus on developing multimodal extraction models to incorporate critical material property and characterization data into future versions of MatSyn Data set, thereby enabling a more comprehensive, data-driven link between synthesis processes and material performance. Meanwhile, it should be recognized that there are some key implicit parameters that are rarely reported in the literature. Future research should capture these contextual experimental factors through enhanced reporting standards and multisource data integration, moving toward more reliable and repeatable intelligent synthesis.

EXPERIMENTAL SECTION

Materials and Reagents

$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and NH_4Cl were brought from Aladdin Chemical Reagent Company (Shanghai, China). All aqueous solutions were prepared using ultrapure water (resistivity = 18.2 M Ω -cm, Milli-Q water) throughout the whole experiment. Nickel foam (NF, 1 mm thickness) was supplied by Shengshijiangxin New Materials Co., Ltd. (Jiangsu, China). Graphite plates were obtained from the High Purity Metal Materials Research Institute (Hefei, China).

Material Synthesis

Preparation of coupled Ni nanocrystals and $\text{Ni}(\text{OH})_2$ materials: Clean the nickel foam in sequence with 1 M HCl solution, ethanol, and deionized water. Dissolve 30 mM $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$, 60 mM $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, and 55 mM NH_4Cl in deionized water to prepare the electrolyte. Assemble the electrodeposition device with graphite as the anode and nickel foam as the cathode. Electrodeposition was performed at a current density of 0.2 A/cm² for 300 s under room temperature and normal pressure. Rinse the deposited material with deionized water and dry it at 60 °C for 2 h.

Characterizations

X-ray diffraction (XRD) analysis was conducted on an X-Pert³ powder diffractometer (PANalytical, Holland) with Cu-K α radiation (λ = 1.5416 Å). For X-ray photoelectron spectroscopy (XPS) measurements, an AXIS Supra system (Kratos Analytical Ltd., England) was utilized, equipped with monochromatic 150 W Al-K α radiation. Raman spectra were recorded with a HORIBA LabRAM HR Evolution Raman spectrometer (Japan) using a 532 nm laser.

ASSOCIATED CONTENT

Data Availability Statement

The data for MatSyn25 is provided at <https://huggingface.co/datasets/MatSynAI/MatSyn25> and <https://github.com/MatSynAI/MatSyn25>. The web platform of MatSyn25 and MatSyn AI is provided at <https://matsynai.stpaper.cn/>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c19849>.

Additional methodology details, statistical tables, hyperparameter analyses, benchmarking results, hallucination assessment data, full model response examples, and XRD, XPS, and Raman characterization for the synthesized material (PDF)

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Author Contributions

[#]C.L. and W.Y. contributed equally to this work. N.R. and L.Q. designed the project. C.L. and Y.W. completed the construction of the project code framework. C.L., Q.W. and Y.L. performed data extraction and processing. Y.W., Z.T. and H.J. performed model training and platform construction. C.L. and Y.W. completed the manuscript. N.R., L.Q., J.L. and Z.Z. revised the manuscript. N.R., L.Q. and J.L. supervised the project and provided funding.

Notes

The authors declare no competing financial interest.

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